Preliminary on Symplectic Varieties and Lagrangian Fibration

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1 Symplectic varieties

Def 1.1. A normal quasi-projective variety V is called *symplectic* if the smooth locus V^{sm} admits a holomorphic symplectic form σ (in particular, $d\sigma = 0$) which extends to a holomorphic form on any resolution of V.



As a caveat when reading literature, this definition is far from the so-called "*irreducible symplectic variety*" as in Matsushita's earlier papers such as [Mat97] which requires V to be compact, simply connected, and $H^0(V, \Omega_V^2)$ spanned by a nowhere degenerate 2-form σ. Though seemingly only differing from irreducibility in the name, the latter is much stronger, and in fact Matsushita showed that every fibration $V \to B$ where V is irreducible symplectic is Lagrangian in [Mat97] using the Beauville-Fujiki form (which we would list as an example later).

Next we shall state the basic property of symplectic variety concerning the singularity. For the sake of completeness we include the definition of some types of singularities.

Def 1.2. Let V be a variety over a field of characteristic 0 and $f : X \to V$ a resolution of singularities. We say that $f : X \to V$ is a *rational resolution* if $f_* \mathcal{O}_X = \mathcal{O}_V$, and $R^i f_* \mathcal{O}_X = 0$ for $i > 0$. We say that V has *rational singularities* if every resolution $f : X \to V$ is rational.

V is said to have *Gorenstein singularities* if it is Cohen-Macaulay and its canonical divisor K_V is Cartier (locally free of rank 1).

Now suppose V is a normal variety such that K_V is $\mathbb{Q}$-Cartier. Let $f : X \to V$ be a resolution of singularities. V has *canonical singularities* if for every exceptional divisor E_i, the discrepancy $a_i \geq 0$.

Prop 1.1. (1) For a symplectic variety V, $\omega_V \cong \mathcal{O}_V$, where ω_V denotes the dualizing sheaf.
(2) A symplectic variety has rational Gorenstein singularities.

Proof. (1) Suppose $\dim V = 2r$ (as algebraic variety) then σ^r (on V^{sm}) is a non-vanishing holomorphic form generating $\omega_{V^{sm}}$ hence $\omega_{V^{sm}}$ is trivial, namely, $\omega_{V^{sm}} \cong \mathcal{O}_{V^{sm}}$. Now V being normal, $\operatorname{codim}(V \setminus V^{sm}) \geq 2$, we deduce $\mathcal{O}_V \cong \omega_V$ from the fact that ω_V is reflexive, and that for a reflexive sheaf $\mathcal{F}$ we have $\mathcal{F} \cong j_*(j^* \mathcal{F})$ for open

immersion $j : U \hookrightarrow X$ with $\mathrm{codim}(X \setminus U) \geq 2$. (Refer to [Har 80] Proposition 1.6). The same also holds for $\mathcal{O}_V$ (which may also be proved more directly using algebraic Hartogs), hence $\omega_V \cong \mathcal{O}_V$.

(2) By (1), the dualizing sheaf is trivial, hence the canonical divisor is 0 and in particular Cartier. Again, consider σ^r and a resolution $\rho : X \to V$, and in particular, $\rho^*\sigma^r$ extends holomorphically. So the discrepancies a_i in the formula $K_X = \rho^* K_V + \sum a_i E_i$ satisfies $a_i \geq 0$ since $\rho^*\sigma^r$ has no poles along the exceptional divisors E_i. Hence the singularities are canonical. so we deduce that the singularities are rational by [KM98] Theorem 5.22, thus V itself is Cohen-Macaulay by [KM98] Lemma 5.12, implying the singularities are Gorenstein. □

E.g. 1.1. There are certainly normal varieties carrying a holomorphic symplectic form on the smooth part, but of non-rational singularity.

Consider the surface $X \subset \mathbb{C}^3$ defined by $x^3 + y^3 + z^3 = 0$. Over the smooth part (i.e., except the origin) we have the holomorphic 2-form given by Poincaré residue $\frac{1}{3z^2} dx \wedge dy = \frac{1}{3x^2} dy \wedge dz = \frac{1}{3y^2} dz \wedge dx$. X being a surface, this is automatically non-degenerate and closed.

But if we blow up the origin o, the exceptional divisor is the elliptic curve $E = \{x^3 + y^3 + z^3 = 0\} \subset \mathbb{P}^2$, hence for this blow-up $\rho : \widetilde{X} \to X$, $(R^1\rho_*\mathcal{O}_{\widetilde{X}})_o \cong \Gamma(X, R^1\rho_*\mathcal{O}_{\widetilde{X}}) \cong H^1(\widetilde{X}, \mathcal{O}_{\widetilde{X}}) \cong \mathbb{C} \neq 0$.

(For the calculation above, one geometric way is to consider $\pi : \tilde{\mathbb{C}}^3 \to \mathbb{C}^3$ that is the blow-up of the origin in the ambient space $\mathbb{C}^3$. The exceptional divisor E' of π is isomorphic to $\mathbb{P}^2$. The restriction of π to the strict transform $\tilde{X}$ gives our resolution $\rho : \tilde{X} \to X$. The total transform of X splits as $\pi^* X = \tilde{X} + 3E'$.

Now we look at the short exact sequence

$$0 \to \mathcal{O}_{\tilde{\mathbb{C}}^3}(-\tilde{X}) \to \mathcal{O}_{\tilde{\mathbb{C}}^3} \to \mathcal{O}_{\tilde{X}} \to 0.$$

Taking the long exact sequence of higher direct images $R^i\pi_*$ yields:

$$\ldots \to R^1\pi_*\mathcal{O}_{\tilde{\mathbb{C}}^3} \to R^1\pi_*\mathcal{O}_{\tilde{X}} \to R^2\pi_*\mathcal{O}_{\tilde{\mathbb{C}}^3}(-\tilde{X}) \to R^2\pi_*\mathcal{O}_{\tilde{\mathbb{C}}^3} \to \ldots$$

Since $\mathbb{C}^3$ is smooth and π is the blow-up of a smooth point, $\tilde{\mathbb{C}}^3$ has rational singularities, implying $R^i\pi_*\mathcal{O}_{\tilde{\mathbb{C}}^3} = 0$ for all $i > 0$. The sequence therefore reduces to an isomorphism: $R^1\pi_*\mathcal{O}_{\tilde{X}} \cong R^2\pi_*\mathcal{O}_{\tilde{\mathbb{C}}^3}(-\tilde{X})$. Now substituting $\tilde{X} \sim \pi^* X - 3E'$ and applying the projection formula, we obtain:

$$\begin{aligned} R^2\pi_*\mathcal{O}_{\tilde{\mathbb{C}}^3}(-\tilde{X}) &= R^2\pi_*\left(\mathcal{O}_{\tilde{\mathbb{C}}^3}(3E') \otimes \mathcal{O}_{\tilde{\mathbb{C}}^3}(-\pi^* X)\right). \\ &\cong R^2\pi_*\mathcal{O}_{\tilde{\mathbb{C}}^3}(3E') \otimes \mathcal{O}_{\mathbb{C}^3}(-X) \end{aligned}$$

Since X is a globally defined principal divisor on $\mathbb{C}^3$, the line bundle $\mathcal{O}_{\mathbb{C}^3}(-X)$ is trivial. We are reduced to computing $R^2\pi_*\mathcal{O}_{\tilde{\mathbb{C}}^3}(3E')$, which is a skyscraper sheaf supported at o,. THis can be done by iteratively taking the long exact sequence associated to the following short exact sequence:

$$0 \to \mathcal{O}_{\tilde{\mathbb{C}}^3}((k-1)E') \to \mathcal{O}_{\tilde{\mathbb{C}}^3}(kE') \to \mathcal{O}_{E'}(kE') \to 0.$$

The normal bundle of the exceptional $\mathbb{P}^2$ in the blow-up of a point in a smooth 3-fold is $\mathcal{O}_{\mathbb{P}^2}(-1)$, so $\mathcal{O}_{E'}(kE') \cong \mathcal{O}_{\mathbb{P}^2}(-k)$. We can settle the calculation by noticing that $H^2(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(-n)) = 0$ for $n = 1, 2$ and $H^1(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(-n))$ always vanishes.

Now after the calculation as carried out above, the stalk of $R^2\pi_*\mathcal{O}_{\mathbb{P}^2}(-3)$ is just $H^2(\mathbb{P}^2, \mathcal{O}_{\mathbb{P}^2}(-3)) \cong \mathbb{C}$.)

Rmk 1.1. As a highly nontrivial result, the requirement for a symplectic variety that the 2-form σ extends holomorphically on any resolution can be equivalently altered to that X has rational Gorenstein singularities, which may seem more intuitive, see [Na 01].

2 Lagrangian fibration

Def 2.1 (Lagrangian subvariety)**.** Let X be a symplectic variety with symplectic form σ. An irreducible subvariety $Y \subset X$, not contained in the singular locus of X, is called *isotropic* if $\sigma|_{Y^{sm} \cap X^{sm}}$ vanishes. Note that $\dim Y \leq \frac{1}{2} \dim X$ by elementary linear algebra. Y is called *Lagrangian* if the equality holds.

Def 2.2 (Lagrangian fibration)**.** Let X be a symplectic variety with symplectic form σ, and B be a normal variety. A proper surjective morphism $f : X \to B$ with connected fibers is called a *Lagrangian fibration* if a general fiber F of f is a Lagrangian subvariety of X with respect to σ, i.e. $\dim F = \frac{1}{2} \dim X$ and $\sigma|_{F^{sm} \cap X^{sm}} = 0$.

E.g. 2.1 (Examples of Lagrangian Fibrations)**.** (1) Hitchin fibration (note that the total space is not compact).

(2) A fibration from irreducible holomorphic symplectic manifold:

Def 2.3. An *irreducible holomorphic symplectic manifold* is a simply connected compact Kähler manifold X, such that $H^0(X, \Omega_X^2)$ is generated by an everywhere non-degenerate holomorphic 2-form σ. (such as K3 surfaces, i.e., compact surface with trivial canonical bundle and $h^{1,0} = 0$, and Kummer surface, and two examples of O'Grady).

Thm 2.1 (Matsushita 1997)**.** Let X be an irreducible holomorphic symplectic manifold of dimension $2n$, and $f : X \to B$ a proper surjective morphism to a normal variety B such that $0 < \dim B < \dim X$, and a general fiber of f is connected, then f is a Lagrangian fibration. Here I only state the result as an example without proof. Time permitting, we would prove this in later talks.

(3) a Lagrangian fibration with all singular fibers irreducible but non-reduced:

We construct a quasi-projective symplectic surface admitting a proper surjective Lagrangian fibration over $\mathbb{A}^1_{\mathbb{C}}$, such that the singular fiber consists of a single irreducible component which is everywhere non-reduced.

Let the ambient space be $\mathcal{Y} = \mathbb{P}^1_{[x_0:x_1]} \times \mathbb{P}^1_{[y_0:y_1]} \times \mathbb{A}^1_v$, and let $Q = x_0y_1 - x_1y_0$. The vanishing locus $Q = 0$ cuts out the diagonal $\Delta \cong \mathbb{P}^1$ inside $\mathbb{P}^1 \times \mathbb{P}^1$. Let $H(x, y)$ be a generic bi-homogeneous polynomial of bidegree $(2, 2)$ in the projective coordinates. (In fact, one can check that $H(x, y) = x_0^2y_0^2 + x_1^2y_1^2 + 2x_0^2y_1^2 + 2x_1^2y_0^2$ is an appropriate choice that is "generic" to satisfy all the transversality and smoothness conditions later in the construction.)

Define the total space $X \subset \mathcal{Y}$ as the hypersurface given by the equation: $F := Q^2 - v \cdot H(x, y) = 0$. Being hypersurface defined by homogeneous polynomial, it is singular at points where F and all partial derivatives of F simultaneously vanish. One can calculate that this requires $H = 0, Q = 0$, and for generic choice of H such that $\nabla F \neq 0$ (such as the choice as shown above), $v = 0$ as well, and there are exact 4 singular points for generic H.

We first verify that X admits a symplectic form on its smooth locus. The ambient space $\mathcal{Y}$ is smooth, with canonical bundle:$K_{\mathcal{Y}} \cong \mathcal{O}(-2,-2) \boxtimes \mathcal{O}_{\mathbb{A}^1}$ (where $\mathcal{O}(-2,-2) = \mathcal{O}(-2) \boxtimes \mathcal{O}(-2)$ is the canonical bundle on $\mathbb{P}^1 \times \mathbb{P}^1$). The hypersurface X is the zero locus of a global section of $\mathcal{O}(2,2) \boxtimes \mathcal{O}_{\mathbb{A}^1}$. By the adjunction formula, the canonical bundle of X is:

$$K_X \cong \left(K_{\mathcal{Y}} \otimes \mathcal{O}_{\mathcal{Y}}(X)\right)\Big|_X \cong \left(\mathcal{O}(-2,-2) \otimes \mathcal{O}(2,2)\right)\Big|_X \cong \mathcal{O}_X.$$

Because $\dim X = 2$ and its canonical bundle is trivial, X possesses a global, nowhere-vanishing holomorphic 2-form σ. As long as H is chosen generically (so that X has only isolated singularities, such as the explicit choice given above), σ restricts to a valid symplectic form on the smooth locus X^{sm}.

Now, being a hypersurface in a smooth variety, X is necessarily Cohen-Macaulay, also we have checked explicitly that X is regular in codimension 1, hence it is normal. Also, passing over to complete local ring (or equivalently, viewing in complex analytic category and consider the corresponding local neighborhood), one may see that the singularity of X is of A_1-type, which is rational. Hence X is indeed a symplectic variety.

Now define the projection morphism $f\colon X \to \mathbb{A}^1_v$ by mapping points to their v-coordinate. Since X is a closed subvariety of $\mathbb{P}^1 \times \mathbb{P}^1 \times \mathbb{A}^1$, the projection onto the $\mathbb{A}^1$ factor is a proper morphism.

The generic fibers when $v \neq 0$ are curves. In fact, they are smooth curves of bidegree $(2,2)$ in $\mathbb{P}^1 \times \mathbb{P}^1$, by Bertini Theorem as long as H is generic. (By the genus formula for complete intersections in $\mathbb{P}^1 \times \mathbb{P}^1$: $g = (d_1 - 1)(d_2 - 1) = (2-1)(2-1) = 1$ Thus, the generic fiber is a smooth proper elliptic curve, agreeing with Thm 2.4 below.) The restriction of the 2-form σ to any 1-dimensional fiber is trivially zero for dimension reasons. Thus, the generic fiber is a Lagrangian subvariety.

Now when $v = 0$, the defining equation becomes:

$$Q^2 = (x_0 y_1 - x_1 y_0)^2 = 0$$

This scheme is supported entirely on the diagonal $\Delta \cong \mathbb{P}^1$. There are no other irreducible components. Due to the exponent, the scheme-theoretic fiber is 2Δ. The fiber has exactly one irreducible component, and it carries a multiplicity of 2 everywhere.

Now we turn to the first important property of Lagrangian fibrations. The smooth case is due to Matsushita [Mat00]; for singular symplectic varieties, see [Sch20, Section 2.6] and the discussion after Definition 3.5 in [MM24]. We retain the torsion-freeness argument and use the singular-locus theorem of [Mat15, Theorem 3.1] to handle components contained in the singular locus. The first input is the following general result of Kollár (with its Kähler extension).

Thm 2.2 (Kollár)**.** Let $f : X \to B$ be a proper surjective morphism from a smooth Kähler manifold to a normal variety. Then $R^i f_* \omega_X$ is torsion free.

Thm 2.3 (Matsushita; singular case as in Schwald)**.** Let $f : X \to B$ be a Lagrangian fibration from a symplectic variety to a normal variety B. Then every irreducible component of every fiber, endowed with its reduced structure, is a Lagrangian subvariety of X. In particular, no such component is contained in $X \setminus X^{sm}$, and f is equidimensional.

Proof. Write $\dim X = 2n$, so that $\dim B = n$. Choose a projective resolution $\rho : Y \to X$ which is an isomorphism over X^{sm}, and put $g = f \circ \rho$. Since X is quasi-projective and f is proper, f and g are projective. The extension $\tilde{\sigma} \in H^0(Y, \Omega_Y^2)$ of $\rho^* \sigma$ need not be non-degenerate, so we do not apply the smooth symplectic theorem directly to g.

First we establish torsion-freeness. The rational singularities of X give

$$R\rho_* \mathcal{O}_Y \simeq \mathcal{O}_X.$$

As X and Y are Cohen–Macaulay of the same dimension, Grothendieck duality then gives $R\rho_* \omega_Y \simeq \omega_X$. The two Leray spectral sequences, together with $\omega_X \cong \mathcal{O}_X$, yield

$$R^i g_* \mathcal{O}_Y \cong R^i f_* \mathcal{O}_X,$$
$$R^i g_* \omega_Y \cong R^i f_* \omega_X \cong R^i f_* \mathcal{O}_X.$$

By Kollár's torsion-freeness theorem, $R^i g_* \omega_Y$, and hence both $R^i f_* \mathcal{O}_X$ and $R^i g_* \mathcal{O}_Y$, are torsion free.

We next construct the cohomology class needed in the proof. From this point until the restriction to a compact fiber resolution, coherent cohomology is understood in the analytic topology. Proper comparison identifies the higher direct images just used with their analytifications. Since $d\sigma = 0$, the holomorphic form $d\tilde{\sigma}$ vanishes on the dense open subset $\rho^{-1}(X^{sm})$, and therefore vanishes on all of Y. Consequently

$$\bar{\partial}\, \overline{\tilde{\sigma}} = \overline{\partial \tilde{\sigma}} = 0.$$

The Dolbeault theorem thus defines a class

$$\alpha = [\overline{\tilde{\sigma}}] \in H^2(Y, \mathcal{O}_Y).$$

This construction uses closedness, rather than Hodge symmetry on the possibly non-compact total space.

Let s be the image of α under the Leray edge map

$$u_g : H^2(Y, \mathcal{O}_Y) \longrightarrow H^0(B, R^2 g_* \mathcal{O}_Y).$$

After restricting to a dense open subset of B, the morphism g is smooth with connected fibers, and each fiber meets $\rho^{-1}(X^{sm})$ densely. The Lagrangian property of the general fiber of f implies that $\tilde{\sigma}$ vanishes on these fibers

of g. Cohomology and base change over this open set therefore show that s vanishes there. As $R^2g_*\mathcal{O}_Y$ is torsion free, $s = 0$ on B.

Let $h : W \to Y$ be a morphism from a smooth projective variety such that $g \circ h$ is the constant map with value b. The restriction of α to W factors through

$$H^2(Y, \mathcal{O}_Y) \xrightarrow{u_g} H^0(B, R^2g_*\mathcal{O}_Y) \longrightarrow (R^2g_*\mathcal{O}_Y) \otimes k(b)$$

and then through the natural maps

$$(R^2g_*\mathcal{O}_Y) \otimes k(b) \longrightarrow H^2(Y_b, \mathcal{O}_{Y_b}) \longrightarrow H^2(W, \mathcal{O}_W).$$

Hence $[\overline{h^*\tilde{\sigma}}] = 0$ in $H^2(W, \mathcal{O}_W)$. No base-change isomorphism at the special point b is needed here. Now W is smooth and projective, so Hodge theory on W gives

$$h^*\tilde{\sigma} = 0. \tag{1}$$

Suppose that F is an irreducible component of a fiber of f and that $F \not\subset X \setminus X^{sm}$. Take for W a projective resolution of the strict transform of F in Y. By (1), σ vanishes on $F^{sm} \cap X^{sm}$. Symplectic linear algebra gives $\dim F \leq n$, whereas the fiber dimension theorem gives $\dim F \geq \dim X - \dim B = n$ for every irreducible fiber component. Thus F is Lagrangian.

It remains to exclude components contained in the singular locus. We do this by induction on $\dim X$. If $\dim X = 2$, normality implies that the singular locus is finite, while every fiber component has dimension at least one. The assertion follows from the preceding paragraph.

For the induction step, let S be an irreducible component of $X \setminus X^{sm}$. By the theorem of Matreshka [Mat15, Theorem 3.1], the normalization S^ν is a symplectic variety and the induced morphism to the normalization of $f(S)$ is Lagrangian; one may equivalently use its Stein factorization. The needed symplectic structure comes from [Kal06, Theorem 2.5]. Write $\dim S^\nu = 2m \leq 2n - 2$. The induction hypothesis implies that all fibers of this lower-dimensional Lagrangian fibration are m-dimensional. Since the normalization and the finite map in the Stein factorization are finite, every nonempty fiber of $f|_S$ has dimension $m \leq n - 1$. The zero-dimensional case is immediate.

If a fiber component F of f were contained in S, this would contradict $\dim F \geq n$. Since the singular locus has finitely many irreducible components, no fiber component is contained in it. The preceding argument now proves that every fiber component is Lagrangian, completing the induction. □

Cor 2.1. Let $f : X \to B$ be a Lagrangian fibration where B is smooth, then f is flat.

The corollary directly follows from the following proposition, known as so-called "Miracle Flatness", since we have seen that a symplectic variety X is Cohen-Macaulay in Prop 1.1. (For a proof, see [Nag62].)

Prop 2.1. Let $A \to B$ be a local morphism of noetherian local rings. Assume that A is regular, B is Cohen-Macaulay, and $\dim B = \dim A + \dim B/\mathfrak{m}_A B$, then $A \to B$ is flat.

Next we come to the second property of Lagrangian fibration, whose proof of this property actually comes directly from the lemma.

Thm 2.4. Any smooth fiber of a Lagrangian fibration $f : X \to B$ is an abelian variety.

Lem 2.1. Any compact Kähler manifold Y with trivial tangent bundle is a complex torus.

Proof. We use Albanese morphism $\alpha_Y : Y \to \mathrm{Alb}(Y) = \frac{H^0(Y,\Omega_Y^1)^*}{H_1(Y,\mathbb{Z})}$, defined by $\alpha_Y(y) = (\theta \mapsto \int_{y_0}^{y} \theta) \pmod{H_1(Y,\mathbb{Z})}$ for a fixed basepoint y_0 and $\theta \in H^0(Y,\Omega_Y^1)$.

Then $d\alpha_Y : T_y Y \to T_{\alpha_Y(y)}\mathrm{Alb}(Y) = H^0(Y,\Omega_Y^1)^*$ is simply the evaluating map $v \mapsto (\theta \mapsto \theta(v))$. Now since $\Omega_Y^1 \cong T_Y^*$ is trivial, $H^0(Y,\Omega_Y^1)$ has a basis $\theta_1, ..., \theta_n$ with these θ_i everywhere linearly-independent, where $n = \mathrm{rank}\, T_Y$.

So $d\alpha_Y$ is an isomorphism everywhere, hence α_Y is a local biholomorphism. Now since Y is compact and $\mathrm{Alb}(Y)$ is Hausdorff (as complex analytic spaces), α_Y is automatically proper, and a proper local homeomorphism is a finite covering. And being a finite covering of a complex torus, Y itself is a complex torus as well. (After passing to fundamental group in view of topological Galois correspondence, this is just saying a finite index subgroup of $\mathbb{Z}^{2n}$ is still isomorphic to $\mathbb{Z}^{2n}$ and we are done.) □

Proof of the Theorem. Being a fiber of a proper map, any fiber X_b over a point b is proper and hence X_b is compact (viewed as a complex analytic space). In view of the lemma above we only need to prove that T_{X_b} is trivial (since being a subvariety of a projective variety X, X_b will be an abelian variety as long as it is a complex torus).

Now suppose f is smooth over b. Without loss of singularity we may restrict ourselves to the smooth part of X. We first notice that this implies the conormal bundle $\mathcal{N}_{X_b/X}$ of X_b in X is trivial. (This is seen by pullback the exact sequence $0 \to f^*\Omega_B \to \Omega_X \to \Omega_{X/B} \to 0$ to the fiber X_b (i.e., tensoring with $\mathcal{O}_{X_b}$, which is left exact here since $\Omega_{X/B}$ is locally free) to get

$$0 \to \mathcal{O}_{X_b} \otimes T_{B,b}^* \cong \mathcal{O}_{X_b}^{\oplus n} \to \Omega_X|_{X_b} \to \Omega_{X_b} \to 0,$$

and compare with the conormal bundle exact sequence of the subvariety X_b inside X:

$$0 \to \mathcal{N}_{X_b/X}^* \to \Omega_X|_{X_b} \to \Omega_{X_b} \to 0.)$$

Now we settle the problem with the isomorphism $T_{X_b} \cong \mathcal{N}_{X_b/X}^*$ induced by the symplectic form since it vanishes identically on X_b. □

3 Relative Dimension Bound

As is seen in previous lectures, when the total space is not smooth and we are pushing forward any complex K such as IC-sheaves instead of constant sheaves, to generalize support theorem as in [MS22], we need to require $\tau_{>2c}(Rf_*K) = 0$, where c is the relative dimension of a weak abelian fibration. We have a following geometric criterion for a equidimensional fibration to determine this vanishing.

Prop 3.1 (Relative Dimension Bound)**.** Let $f : X \to B$ be an equidimensional proper morphism of complex algebraic varieties, with fibers of pure dimension c. Use the unshifted normalization $\mathrm{IC}(X, \mathbb{Q})|_{X^{sm}} = \mathbb{Q}_{X^{sm}}$. Suppose that there exists an algebraic Whitney stratification $X = \bigsqcup_i X_i$, adapted to X^{sm} and to the intersection complex, such that whenever $X_i \cap f^{-1}(b) \neq \emptyset$, we have

$$\operatorname{codim}_X X_i \leq 2 \operatorname{codim}_{f^{-1}(b)} (X_i \cap f^{-1}(b)). \tag{2}$$

Then

- $\tau_{>2c}(Rf_* \mathrm{IC}(X, \mathbb{Q})) = 0$;
- the natural map $\mathbb{Q}_X \to \mathrm{IC}(X, \mathbb{Q})$ induces an isomorphism

$$R^{2c} f_* \mathbb{Q}_X \xrightarrow{\sim} \mathscr{H}^{2c}(Rf_* \mathrm{IC}(X, \mathbb{Q})).$$

Here τ and $\mathscr{H}$ refer to the standard, not the perverse, t-structure.

Proof. Fix $b \in B$, and put $F = f^{-1}(b)_{\mathrm{red}}$ and $K = \mathrm{IC}(X, \mathbb{Q})|_F$. Proper base change gives

$$\mathscr{H}^j(Rf_* \mathrm{IC}(X, \mathbb{Q}))_b \cong \mathbb{H}^j(F, K).$$

Consider the hypercohomology spectral sequence

$$E_2^{a,q} = H^a(F, \mathscr{H}^q(K)) \Longrightarrow \mathbb{H}^{a+q}(F, K).$$

For a nonempty intersection with a stratum, write

$$k_i = \operatorname{codim}_X X_i, \qquad d_i = \dim(X_i \cap F).$$

Condition (2) is equivalent to $2d_i \leq 2c - k_i$. The intersection complex has no negative cohomology sheaves. Its strong support condition says that, if $q > 0$ and $\mathscr{H}^q(\mathrm{IC}(X, \mathbb{Q}))$ is nonzero on X_i, then $q \leq k_i - 1$. Thus, on every such stratum,

$$q + 2d_i \leq (k_i - 1) + (2c - k_i) = 2c - 1.$$

Taking the maximum over the strata on which the restricted sheaf is nonzero gives

$$q + 2\dim \operatorname{Supp} \mathscr{H}^q(K) \le 2c - 1 \qquad (q > 0).$$

A constructible sheaf with support of complex dimension d has no cohomology in degrees greater than $2d$. It follows that $E_2^{a,q} = 0$ for $q > 0$ and $a + q \ge 2c$. For $q = 0$, the support has dimension at most c, so $E_2^{a,0} = 0$ for $a > 2c$. This proves $\mathbb{H}^j(F, K) = 0$ for $j > 2c$, hence the first assertion.

For the second assertion, one must also exclude differentials entering the term $E_r^{2c,0}$ from total degree $2c - 1$. We give a weight argument. Set

$$U = F \cap X^{sm}, \qquad Z = F \setminus U.$$

For a stratum contained in the singular locus, $k_i > 0$, so (2) implies $d_i < c$. Consequently $\dim Z < c$ and every irreducible component of F meets U. Applying the preceding support estimate on Z gives

$$\mathbb{H}^{2c}(Z, K|_Z) = 0 :$$

for $q > 0$ the possible total degrees are at most $2c - 1$, and for $q = 0$ they are at most $2\dim Z < 2c$.

Since $K|_U = \mathbb{Q}_U$, the localization triangle yields an exact sequence of mixed Hodge structures

$$\mathbb{H}^{2c-1}(Z, K|_Z) \longrightarrow H_c^{2c}(U, \mathbb{Q}) \longrightarrow \mathbb{H}^{2c}(F, K) \longrightarrow 0.$$

The unshifted intersection complex is pure of weight zero. Pullback preserves upper bounds on weights, and Z is proper, so the first term has weights at most $2c - 1$. The middle term is pure of weight $2c$: if r is the number of irreducible components of F, it is isomorphic to $\mathbb{Q}(-c)^{\oplus r}$. The first arrow must therefore be zero. These weight properties are standard consequences of Saito's theory; see [Sai16]. Hence

$$H_c^{2c}(U, \mathbb{Q}) \xrightarrow{\sim} \mathbb{H}^{2c}(F, K).$$

On the other hand, localization for the constant sheaf and $\dim Z < c$ give

$$H_c^{2c}(U, \mathbb{Q}) \xrightarrow{\sim} H^{2c}(F, \mathbb{Q}).$$

The natural map $\mathbb{Q}_F \to K$ is the identity over U and is compatible with both localization sequences. It therefore induces an isomorphism $H^{2c}(F, \mathbb{Q}) \to \mathbb{H}^{2c}(F, K)$. Applying proper base change at every b proves the asserted isomorphism of sheaves. □

The inequality in (2) is written in the direction needed by the support estimate above; it corrects the direction printed in [MM24, Proposition 3.3, equation (14)].

Prop 3.2. Let X be a symplectic variety endowed with a Lagrangian fibration $f : X \to B$. Then X admits a complex algebraic Whitney stratification

$$X = \bigsqcup_i X_i$$

by locally closed smooth symplectic subvarieties such that every nonempty intersection $X_i \cap f^{-1}(b)$ has pure dimension $\frac{1}{2}\dim X_i$ and is isotropic in X_i. In particular, whenever this intersection is nonempty,

$$\operatorname{codim}_X X_i = 2\operatorname{codim}_{f^{-1}(b)}(X_i \cap f^{-1}(b)).$$

Proof. We use the symplectic Whitney stratification of [MM24, Proposition 3.6], based on Kaledin's stratification theorem [Kal06, Theorem 2.3 and Proposition 3.1]. Its strata are obtained from the smooth loci of the successive symplectic pieces; the relevant normalizations of their closures are symplectic varieties. Applying the theorem of Matreshka [Mat15, Theorem 3.1] successively gives proper Lagrangian fibrations on these normalizations, with Stein factorization if needed. The symplectic forms agree with the induced forms on the open strata.

By Theorem 2.3, the fibers on each normalization have pure dimension equal to half the dimension of that normalization and are isotropic. Restricting to the corresponding open stratum gives the stated assertion for $X_i \cap f^{-1}(b)$. Notice that $f|_{X_i}$ itself need not be proper; the proper fibrations used here are those on the normalizations of the closed pieces.

Theorem 2.3 also gives $\dim f^{-1}(b) = \frac{1}{2}\dim X$. Therefore, for every nonempty intersection,

$$\begin{aligned}\operatorname{codim}_X X_i &= \dim X - \dim X_i \\ &= 2\dim f^{-1}(b) - 2\dim(X_i \cap f^{-1}(b)) \\ &= 2\operatorname{codim}_{f^{-1}(b)}(X_i \cap f^{-1}(b)).\end{aligned}$$

□

References for the revised arguments

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